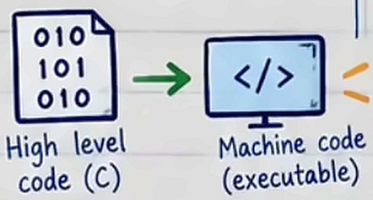
Wipro Interview Technical Notes

Code Learn Grow Placement ... ☺

1 What is the difference between compiler and interpreter ?

Compiler	Interpreter
• Translates the entire program at once.	• Translates line by line.
• Generates machine code (executable file).	• Does not generate executable file.
• Faster execution.	• Slower execution.
• Shows all errors after compilation.	• Stops at the first error.
• Example: GCC (C compiler)	• Example: Python interpreter



2 What are variables and data types in C ?

- **Variable** is a name given to a memory location to store data which can change during program execution.
- **Data type** specifies the type of data a variable can hold.

Common data types:

- **int** → stores integers (e.g., 10, -5)
- **float** → stores decimal numbers (e.g., 3.14)
- **char** → stores single character (e.g., 'A')
- **double** → stores large decimal numbers (e.g., 3.14159)

3 What is the difference between == and = ?

- **=** is an assignment operator. It assigns a value to a variable.
- **==** is an equality operator. It checks whether two values are equal or not.

Example:

```
int a = 5;    // assignment
if (a == 5) { // comparison
    printf("Equal");
}
```

a = 5 → store 5 in variable a

a == 5 → check if a is equal to 5

4 What are loops in C ?

- Loops are used to execute a block of code repeatedly until a condition is false.
- **Types of loops in C:**
  - **for loop** → used when number of iterations is known.
  - **while loop** → used when number of iterations is not known (condition based).
  - **do-while loop** → executes at least once, then checks the condition.

Repeat until condition is false

5 What is a function in C ?

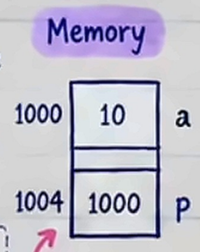
- A function is a reusable block of code that performs a specific task.
- It can take input values (parameters) and can return a value (return type).
- Helps in modular programming and code reusability.

Parameters make function flexible! ☺

```
// function to add two numbers
int add(int a, int b) {    // return type: int
    return a + b;
}
```

6 What is a pointer ?

- A pointer is a variable that stores the address of another variable.
- It helps in dynamic memory allocation and efficient memory usage.




```
int a = 10;
int *p = &a; // p stores address of a
printf("Value: %d", a); // 10
printf("Address: %u", p); // address of a
```

p stores address of a (1000)

Tip: explain with small examples ☺

Wipro Interview Technical Notes

Code Learn Grow Placement ... ☺

1 What is OOP ?

- OOP (Object Oriented Programming) is a programming paradigm that organizes code using objects and classes.
- It helps in writing modular, reusable and maintainable code.
- Real world problems are modeled using objects (e.g., Car, Student, Employee).

Example:

```
class Student {
    String name;   // class with
    int age;       // data (variables)
}
```

Class (Student) → creates → Object (s1, s2, ...)

2 Four pillars of OOP

- The four main features (pillars) of OOP are:

1. **Encapsulation** → Hides data using access modifiers.
2. **Abstraction** → Shows only essential features and hides implementation.
3. **Inheritance** → Allows a new class to inherit properties of an existing class.
4. **Polymorphism** → Allows one interface to have multiple forms (many behaviors).

These four pillars help in achieving reusability, security and flexibility in code.

3 Difference between class and object

Class	Object
• Blueprint or template for creating objects.	• Instance of a class.
• Does not occupy memory.	• Occupies memory.
• Defines properties and methods.	• Uses the properties and methods.
• Only one class can exist.	• Multiple objects can be created from a class.

Example:
`class Car {`
`  String brand;`
`}`
	Example:
	`Car c1 = new Car();`
	`c1.brand = "Toyota";`
	`// c1 is an object`

4 Difference between overloading and overriding

Method Overloading	Method Overriding
• Same method name with different parameters.	• Same method name and same parameters (but different implementation).
• Occurs in the same class.	• Occurs in parent-child classes (inheritance).
• Compile-time polymorphism (static binding).	• Runtime polymorphism (dynamic binding).

Example:
`class Math {`
`  int add(int a, int b){`
`  return a + b;`
`  }`
`  int add(int a, int b, int c){`
`  return a + b + c;`
`  }`
`}`
	Example:
	`class Animal {`
	`  void sound() {`
	`  System.out.println("Animal");`
	`  }`
	`}`
	`class Dog extends Animal {`
	`  void sound() { // overriding`
	`  System.out.println("Bark");`
	`}`

Same name Different behavior :)

5 What is constructor ?

- A constructor is a special method that is called automatically when an object is created.
- It is used to initialize the object's data members,
- Constructor name is same as the class name and it does not have a return type.

Example:

```
class Student {
    String name;
    Student(String n) {    // constructor
        name = n;
    }
}
```

Constructor is called automatically when object is created :)

6 What is abstraction vs encapsulation ?

Abstraction	Encapsulation
• Hides implementation details and shows only essential features (what to do).	• Hides data (variables) by using access modifiers and provides controlled access using getters and setters (how to do).
• Achieved using abstract classes and interfaces.	• Bundles data and methods to protect data from direct access.
• **Example:** Shape (draw()) – we don't know how it draws, only call draw().	• **Example:** private variables with public getter/setter methods.

Tip: OOP concepts are frequently asked in interviews. Be clear with examples :)

Wipro Interview Technical Notes



1 What is an array ?

- An array is a collection of elements of the same data type stored in contiguous memory locations.
- Elements are accessed using an index (starting from 0).
- Size of array is fixed (in most languages like C).

Example: Array of 5 integers

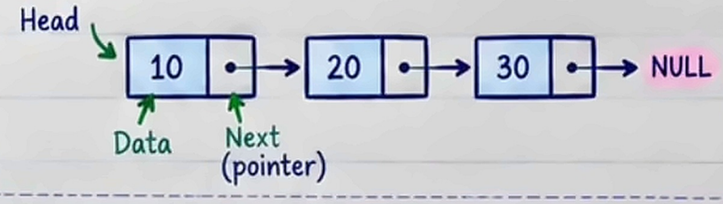



```
int arr[5] = {10, 20, 30, 40, 50};
```

2 What is a linked list ?

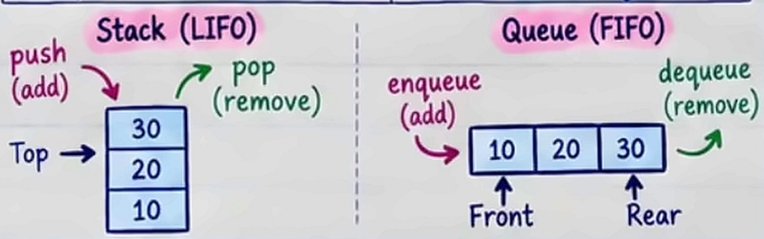
- A linked list is a linear data structure where elements (nodes) are stored at non-contiguous memory locations.
- Each node contains data and a reference (pointer) to the next node.
- It allows dynamic size (can grow or shrink during execution).

Example: Singly Linked List



3 Difference between stack and queue ?

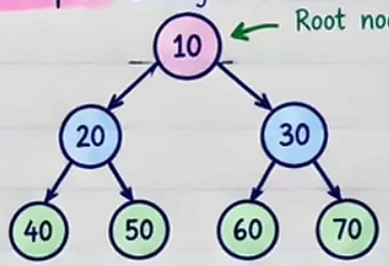
Stack	Queue
• Follows LIFO (Last In First Out).	• Follows FIFO (First In First Out).
• Insert and delete from the same end (top).	• Insert at rear and delete from front.
• Operations: push(), pop(), top().	• Operations: enqueue(), dequeue(), front().
• Example: Undo operation, function calls.	• Example: Print queue, CPU scheduling.



4 What is a tree ?

- A tree is a hierarchical non-linear data structure consisting of nodes connected by edges.
- It has a root node, and each node can have zero or more child nodes.
- There are no cycles in a tree.
- Common types: Binary Tree, Binary Search Tree, AVL Tree, etc.

Example: Binary Tree



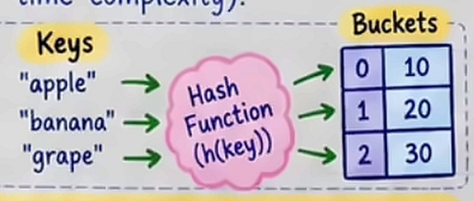
Each node can have at most two children (in a binary tree)

5 What is a hash table ?

- A hash table (or hash map) stores key-value pairs using a hash function.
- The hash function converts a key into an index (bucket) in an array.
- It allows fast insertion, deletion and search on average (O(1) time complexity).

Example:

```
hash["apple"] = 10;
hash["banana"] = 20;
hash["grape"] = 30;
```



6 Difference between linear and non-linear data structures ?

Linear Data Structures	Non-Linear Data Structures
• Elements are arranged in a sequential order.	• Elements are arranged in a hierarchical or network structure.
• Each element (except first and last) has a single predecessor and successor.	• A node can have multiple predecessors and successors.
• Examples: Array, Linked List, Stack, Queue.	• Examples: Tree, Graph, Heap.

Tip: Practice drawing diagrams for each data structure :)

Wipro Interview Technical Notes

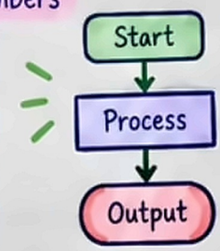


1 What is an algorithm ?

- An algorithm is a step-by-step procedure to solve a problem.
- It takes input, performs a set of instructions and produces output.
- It should be finite, well-defined and give correct result.

Example: Find sum of first n numbers

1. Start
2. sum = 0
3. For i = 1 to n, add i to sum
4. Return sum
5. End



2 What is time complexity ?

- Time complexity measures how the running time of an algorithm grows with input size (n).
- It helps us compare the efficiency of algorithms.
- We use Big-O notation to represent the growth rate.



Example: Sum of n numbers

```
int sum = 0;
for (int i = 1; i <= n; i++) {
    sum += i;
}
```

This loop runs n times, so time complexity is O(n).

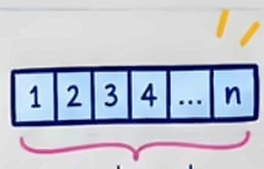
3 What is space complexity ?

- Space complexity measures the amount of memory used by an algorithm.
- It includes memory for variables, data structures and recursive calls.
- We also use Big-O notation for space complexity.

Example: Array of size n

```
int arr[n]; // requires memory
            // for n integers
```

Space complexity = O(n)



4 Common Big-O values

Big-O	Name	Example / When it occurs
O(1)	Constant	Accessing an array element arr[i]
O(log n)	Logarithmic	Binary search (divide and conquer)
O(n)	Linear	Traversing an array
O(n log n)	Linearithmic	Sorting algorithms (Merge Sort, Heap Sort)
$O(n^2)$	Quadratic	Nested loops (for each element, loop again)

5 Difference between linear search and binary search

Linear Search	Binary Search
• Works on unsorted or sorted array.	• Works only on sorted array.
• Checks each element one by one.	• Divides the search space in half each time.
• Time complexity: O(n)	• Time complexity: O(log n)
• Simple to implement.	• More efficient than linear search.

Example: Search for 7 in array: [2, 4, 7, 8, 10]

Linear Search: 2 → 4 → 7 (3 comparisons)

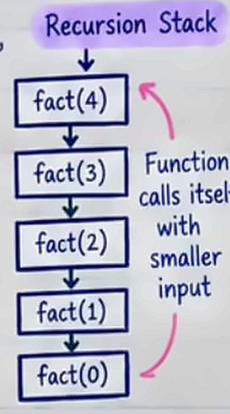
Binary Search: Check middle (7) (1 comparison)

6 What is recursion ?

- Recursion is a technique where a function calls itself to solve a smaller subproblem.
- It has a base case to stop the recursion.
- It is useful for problems like factorial, fibonacci, tree traversal etc.

Example: Factorial using recursion

```
int fact(int n) {
    if (n == 0) return 1;  // base case
    return n * fact(n - 1);
}
```



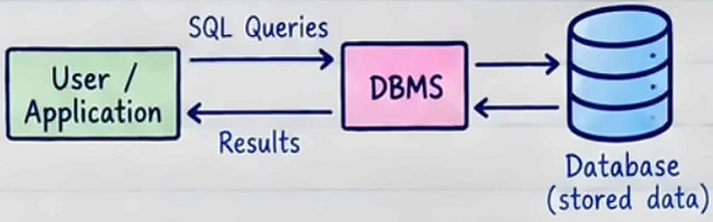
Tip: In interviews, always mention the time and space complexity of your solution.

Wipro Interview Technical Notes



1 What is DBMS ?

- DBMS (Database Management System) is software used to store, manage, retrieve and organize data efficiently.
- It provides an interface between users and the data stored in the database.
- **Examples:** MySQL, Oracle, Microsoft SQL Server, PostgreSQL.



2 Difference between DBMS and RDBMS ?

Feature	DBMS	RDBMS
Data model	Can be hierarchical, network, file-based or relational.	Relational model (table based).
Data storage	May or may not use tables.	Uses tables (rows and columns).
Relationships	Limited support for relationships.	Supports relationships using keys (PK, FK).
Consistency	Basic level.	Better data consistency.
Examples	File system, older systems.	MySQL, Oracle, PostgreSQL, SQL Server.

3 What is normalization ?

- Normalization is the process of organizing data in a database to reduce redundancy and improve data integrity.
- It splits large tables into smaller related tables.
- Common normal forms: 1NF, 2NF, 3NF, BCNF.

Example (before normalization):

StudentID	Name	Course1	Course2
1	Rahul	DBMS	OS
2	Neha	DBMS	CN

↓ After normalization (separate tables):

Student	
StudentID	Name
1	Rahul
2	Neha

Course	
StudentID	Course
1	DBMS
1	OS
2	DBMS
2	CN

4 What is primary key and foreign key ?

- **Primary Key:** A unique identifier for each record in a table. It cannot be NULL and does not allow duplicate values.
- **Foreign Key:** A field in one table that refers to the primary key of another table. It creates a relationship between tables.

Example:

StudentID (FK) refers to StudentID (PK)

Student	
StudentID (PK)	Name
1	Rahul
2	Neha
3	Karan

Enroll		
EnrollID	StudentID (FK)	Course
101	1	DBMS
102	2	OS
103	1	CN

5 What is ACID property ?

- ACID properties ensure reliable processing of database transactions.



- **A – Atomicity** → All or nothing.
- **C – Consistency** → Database remains in a consistent state.
- **I – Isolation** → Transactions do not interfere with each other.
- **D – Durability** → Changes are permanent even after a failure.

6 What is transaction ?

- A transaction is a sequence of operations performed as a single logical unit of work.
- It follows ACID properties.
- **Example:** Transfer money from one account to another.

```
BEGIN TRANSACTION;
UPDATE Account SET balance = balance - 100
 WHERE acc_id = 1;
UPDATE Account SET balance = balance + 100
 WHERE acc_id = 2;
COMMIT;
```



Transferred successfully (as one transaction)

Tip: Practice basic SQL queries (SELECT, INSERT, UPDATE, DELETE) for interviews ;)

Wipro Interview Technical Notes

Page 6/10 • SQL Queries & Joins



Query
Filter
Join
Aggregate
Practice

1 What is SQL ?

- SQL (Structured Query Language) is used to store, manage and retrieve data from relational databases.
- It is used to perform operations like CREATE, READ, UPDATE and DELETE data.
- SQL works with tables (rows and columns).
- Commonly used in interview questions for data retrieval and joins.

Example:

```
SELECT *
FROM employees;
```

employees		
id	name	dept
1	Alice	HR
2	Bob	IT
3	Carol	HR

2 Difference between WHERE and HAVING

WHERE	HAVING
• Used to filter rows before grouping.	• Used to filter groups after grouping.
• Works on individual rows.	• Works on aggregated results (GROUP BY).
• Cannot be used with aggregate functions like COUNT(), SUM().	• Can be used with aggregate functions.

Example:

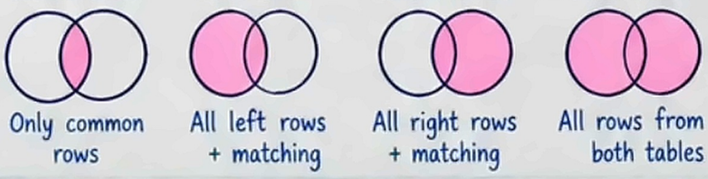
```
-- WHERE (row level filter)
SELECT * FROM employees
WHERE dept = 'HR';
```

```
-- HAVING (group filter)
SELECT dept, COUNT(*)
FROM employees
GROUP BY dept
HAVING COUNT(*) > 2;
```

3 Types of JOINs in SQL

- JOIN is used to combine rows from two or more tables based on a related column.
- Common types of JOINs:

INNER JOIN → returns only matching rows.
LEFT JOIN → all rows from left table.
RIGHT JOIN → all rows from right table.
FULL JOIN → all rows from both tables.



4 Difference between DELETE, DROP and TRUNCATE

Feature	DELETE	TRUNCATE	DROP
Purpose	Deletes specific rows or all rows (using condition).	Removes all rows from a table.	Deletes the entire table structure.
WHERE clause	Yes	No	No
Table structure	Remains	Remains	Also removed
Speed	Slower	Faster	Faster
Rollback possible	Yes	No (in most DBs)	No

Example:

```
DELETE FROM employees
WHERE dept = 'HR';
```

```
TRUNCATE TABLE
employees;
```

```
DROP TABLE
employees;
```

5 GROUP BY and ORDER BY

- GROUP BY is used to group rows that have same values in a column, often with aggregate functions (COUNT, SUM, AVG etc).
- ORDER BY is used to sort the result set in ascending (ASC) or descending (DESC) order.

Example (GROUP BY):

```
SELECT dept, COUNT(*)
FROM employees
GROUP BY dept;
```

Example (ORDER BY):

```
SELECT * FROM employees
ORDER BY salary DESC;
```

6 Simple SELECT query & Aggregate functions

- SELECT is used to retrieve data from a table.
- Aggregate functions perform calculations on a set of values, such as COUNT(), SUM(), AVG(), etc.

Example:

```
SELECT id, name, dept, salary
FROM employees
WHERE salary > 50000
ORDER BY salary DESC;
```

employees			
id	name	dept	salary
1	Alice	HR	60000
2	Bob	IT	45000
3	Carol	HR	50000
4	David	IT	70000

Aggregate Functions:

- COUNT() → returns number of rows.
- SUM() → returns sum of a column.
- AVG() → returns average of a column.

Example:

```
SELECT COUNT(*) FROM employees;
SELECT SUM(salary) FROM employees;
SELECT AVG(salary) FROM employees;
```

Tip: practice writing queries on a sample table ;)

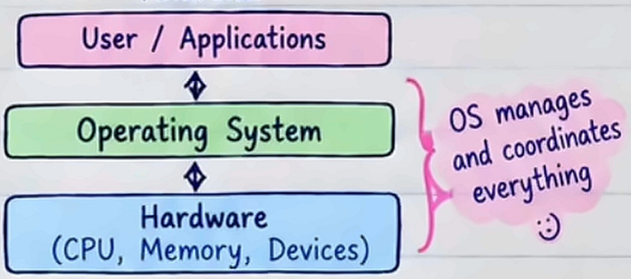
Wipro Interview Technical Notes



Code
Learn
Grow
Placement
... ☺

1 What is an operating system ?

- An Operating System (OS) is system software that acts as an interface between user and hardware.
- It manages system resources like CPU, memory, files, and devices.
- Provides services for programs to run.
- **Examples:** Windows, Linux, macOS, Android.

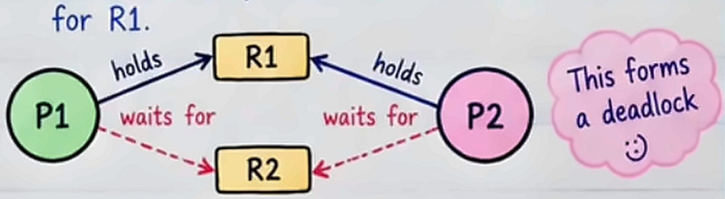


2 Difference between process and thread ?

Aspect	Process	Thread
Definition	A program in execution.	Smallest unit of execution within a process.
Memory	Has its own separate memory.	Shares memory with other threads of same process.
Creation	Slower (heavyweight)	Faster (lightweight)
Communication	Uses IPC (inter-process communication).	Shares data easily (same address space).
Example	Chrome, VS Code (if opened separately)	Multiple threads inside a web browser.

3 What is deadlock ?

- Deadlock is a situation where two or more processes are waiting for each other to release resources, and none of them can proceed.
- It leads to a state where the processes are blocked forever.
- **Example:** Process P1 holds resource R1 and waits for R2, while P2 holds R2 and waits for R1.

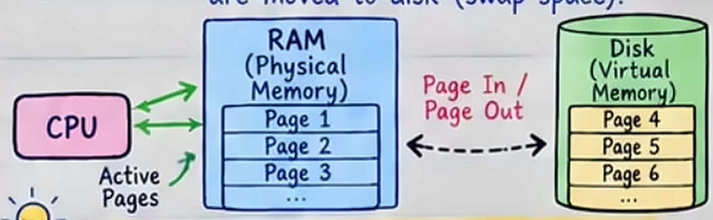


4 Difference between RAM and ROM ?

Aspect	RAM	ROM
Full Form	Random Access Memory	Read Only Memory
Nature	Volatile (data lost when power is off)	Non-volatile (data retains even without power)
Usage	Stores currently running programs and data.	Stores firmware and boot instructions (e.g., BIOS)
Read/Write	Read and Write	Mostly Read Only
Speed	Faster	Slower
Example	DRAM, SRAM	PROM, EPROM, EEPROM

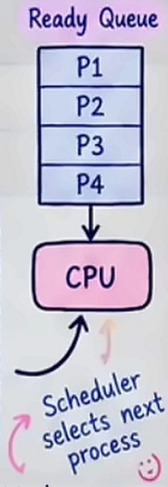
5 What is virtual memory ?

- Virtual memory is a memory management technique that uses a portion of hard disk as an extension of RAM.
- It allows running larger programs than the physical memory size.
- It uses paging to manage data between RAM and disk.
- **Example:** When RAM is full, inactive pages are moved to disk (swap space).



6 What is CPU scheduling ?

- CPU scheduling is the process of selecting a process from the ready queue to execute on the CPU.
- The goal is to maximize CPU utilization and minimize waiting time.
- Types of scheduling algorithms:
  - **FCFS** → First Come First Serve (simple but can cause long waiting time)
  - **SJF** → Shortest Job First (minimizes waiting time)
  - **Priority Scheduling** → higher priority process executes first
  - **Round Robin** → each process gets a fixed time quantum (fair scheduling)



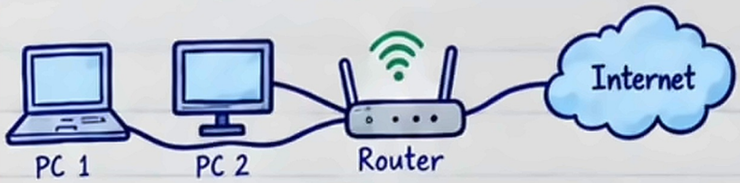
Tip: practice OS diagrams and real world examples ☺

Wipro Interview Technical Notes

Code Learn Grow Placement ...

1 What is a computer network ?

- A computer network is a collection of interconnected devices that share resources and communicate with each other using standard protocols.
- It can share data, files, printers, internet connection, etc.
- **Example:** Multiple computers in an office connected through a router to access the internet.



2 What is IP address ?

- An IP address (Internet Protocol address) is a unique numerical address given to each device in a network.
- It helps in identifying the device and routing data to it.
- Two types: IPv4 and IPv6

Example:

- IPv4: 192.168.1.10 (32-bit, written in dotted decimal format)
- IPv6: 2001:0db8:85a3:0000:0000:8a2e:0370:7334 (128-bit)

IP address identifies your device on the network :)



3 Difference between TCP and UDP ?

Feature	TCP	UDP
Connection	Connection-oriented	Connectionless
Reliability	Reliable (guarantees delivery)	Not reliable (no guarantee)
Speed	Slower	Faster
Data order	Maintains order	Does not maintain order
Use cases	Web browsing, file transfer, email (HTTP, FTP, SMTP)	Video streaming, online gaming, DNS, VoIP
Example	TCP (port 80, 443)	UDP (port 53, 67)

4 What is OSI model ?

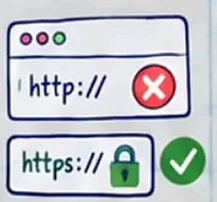
- The OSI (Open Systems Interconnection) model is a 7-layer conceptual model that defines how data is transmitted between two devices over a network.
- It helps in understanding and standardizing the communication process.

7	Application	→ Provides network services to user
6	Presentation	→ Data format, encryption, compression
5	Session	→ Manages sessions between applications
4	Transport	→ End-to-end communication (TCP/UDP)
3	Network	→ Routing and logical addressing (IP)
2	Data Link	→ Framing, MAC addressing (Ethernet)
1	Physical	→ Physical connection, transmission of bits

Data flows from Layer 7 to Layer 1 (sending) and 1 to 7 (receiving) :)

5 What is HTTP vs HTTPS ?

- **HTTP** (HyperText Transfer Protocol) is used to transfer data between a web browser and a web server.
- **HTTPS** (HTTP Secure) is the secure version of HTTP. It uses SSL/TLS encryption to protect data from being intercepted.



Feature	HTTP	HTTPS
Security	Not secure (data in plain text)	Secure (data encrypted)
Port number	80	443
URL format	http://example.com	https://example.com
Use cases	Non-sensitive data	Sensitive data (login, payments)

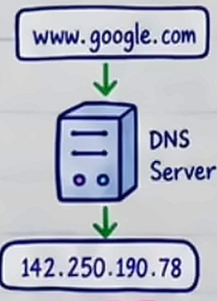
HTTPS keeps your data safe using encryption :)

6 What is DNS ?

- **DNS** (Domain Name System) is like a phonebook for the internet. It translates human-readable domain names into IP addresses.
- It helps us access websites using names like www.google.com instead of IP addresses.

Example:

- You type www.google.com in browser.
- DNS translates it to an IP address (e.g., 142.250.190.78).
- Then your device connects to that IP address to load the website.



DNS converts names → IP addresses :)

Wipro Interview Technical Notes



1 What is HTML ?

- HTML (HyperText Markup Language) is used to create the structure of a web page.
- It defines elements like headings, paragraphs, links, images, etc.
- It uses tags enclosed in < >.



Example:

```
<h1>Welcome to Wipro</h1>
<p>This is a web page.</p>
<a href="https://wipro.com">Visit</a>
```

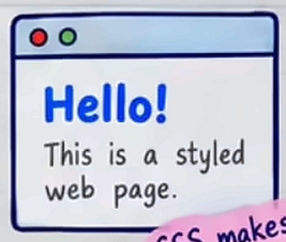
HTML gives structure

2 What is CSS ?

- CSS (Cascading Style Sheets) is used to style and design web pages.
- It controls the layout, colors, fonts, spacing, etc.
- It makes the web page look attractive.

Example:

```
h1 {
    color: blue;
    font-size: 24px;
}
p {
    color: gray;
}
```



CSS makes it look good

3 What is JavaScript ?

- JavaScript is a programming language used to make web pages interactive.
- It runs in the browser and can handle events, manipulate the DOM, and fetch data from APIs.
- It is used along with HTML and CSS.

Example:

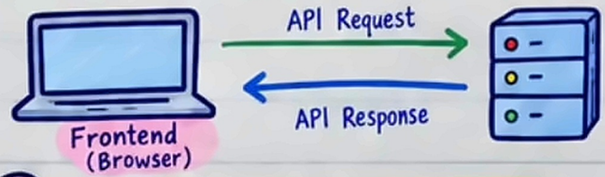
```
<button onclick="greet()">
  Click Me</button>
<script>
  function greet() {
    alert("Hello, Wipro!");
  }
</script>
```

JavaScript adds interactivity



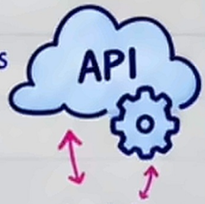
4 Difference between frontend and backend ?

Frontend (Client Side)	Backend (Server Side)
• Runs in the browser.	• Runs on the server.
• Handles user interface and user experience.	• Handles business logic, database operations.
• Built using HTML, CSS, JavaScript.	• Built using languages like Java, Python, Node.js, etc.
• Interacts with backend using APIs.	• Provides APIs to the frontend.
• Example: React, Angular, Vue.js	• Example: Node.js, Spring Boot, Django



5 What is API ?

- API (Application Programming Interface) allows different software applications to communicate with each other.
- It defines a set of rules and endpoints to send and receive data.
- Commonly used over HTTP/HTTPS.



Acts as a bridge between client and server

Example: (GET Request)

```
fetch("https://api.example.com/users")
  .then(response => response.json())
  .then(data => console.log(data));
```

6 What is cloud computing ?

- Cloud computing is the delivery of computing resources (servers, storage, databases, etc.) over the internet.
- It helps in scalable, flexible and cost-effective solutions.
- Widely used by companies for deploying and managing applications.



Examples of cloud platforms:

- Amazon Web Services (AWS)
- Microsoft Azure
- Google Cloud Platform (GCP)

Access resources over the internet anytime, anywhere



Tip: Revise basic terms and try to build a small project using HTML, CSS and JavaScript :)

Wipro Interview Technical Notes

Page 10/10 • Software Engineering Basics

Code Learn Grow Placement ...

1 What is SDLC ?

- SDLC (Software Development Life Cycle) is a process followed to develop high quality software in a structured way.
- It defines different stages from requirement gathering to maintenance.

Phases of SDLC:

Requirement Analysis → Design → Implementation (Coding) → Testing → Deployment → Maintenance

Goal: deliver reliable, maintainable and user-friendly software.

2 Agile vs Waterfall ?

- Agile and Waterfall are two different approaches to software development.
- Agile is iterative and flexible, while Waterfall is linear and sequential.

Aspect	Agile	Waterfall
Approach	Iterative & incremental	Linear & sequential
Requirement	Can change	Fixed at start
Feedback	Regular feedback	Feedback at end
Delivery	Small increments	Single final release
Best suited for	Dynamic projects	Well-defined projects

Example: Agile → Sprint (2-4 weeks) → Deliver feature
Waterfall → Complete one phase → Move to next

3 What is version control / Git ?

- Version control system (VCS) helps to track changes in code, collaborate with others, and maintain history.
- Git is a distributed version control system.
- It allows multiple developers to work on the same project without overwriting each other's code.

Common Git commands:

- git init → initialize repository
- git clone <url> → clone repository
- git add . → stage changes
- git commit -m "msg" → commit changes
- git push → push to remote
- git pull → fetch and merge changes



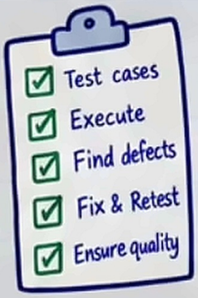
Helps in team collaboration :)

4 What is testing ?

- Testing is the process of finding defects in a software application to ensure it works as expected.
- It helps in delivering a reliable and bug-free product.

Types of testing:

- **Unit Testing** → tests individual functions or modules.
- **Integration Testing** → tests interaction between modules.
- **System Testing** → tests the complete application as a whole.



Example: For a login feature:

- Unit test → test validateUser() function.
- Integration test → test with database.
- System test → test complete login flow (UI + Backend).

5 What is debugging ?

- Debugging is the process of finding and fixing errors (bugs) in the code.
- It involves identifying the root cause of an issue and correcting it.



Find the bug & Fix it!

Example:

```
int a = 5, b = 0;
printf("%d", a / b); // causes runtime error
// Fix: check for zero before division
if (b != 0) {
   printf("%d", a / b);
}
```

6 What is API testing or Black-box vs White-box ?

- API testing checks whether APIs work correctly and return the expected response.
- Black-box and White-box are two different testing techniques.



API Testing:

- Send request to API endpoint.
- Verify response status code, body and headers.

```
// Example using curl
curl -X GET "https://api.example.com/users"
// Expected: 200 OK with JSON response
```

Black-box	White-box
• Tests functionality without knowing internal code. • Focus on inputs and outputs. • **Example:** API testing, UI testing.	• Tests internal code and logic. • Focus on code coverage. • **Example:** unit testing, branch testing.

Tip: Practice small projects and apply these concepts. :)